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EDS
CodeTantra Codes

Name: Aniket A.D. Avdhut
PRN: 202501040048
Division: CS1
Batch: C13
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Practical 1

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1.1.1. Calculate Momentum

09:39

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:
m is the mass of the object (in kilograms).
v is the velocity of the object (in meters per second).

Input Format:
- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:
- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases +

calculate... Submit

```python
#Type Content here...
m=float(input())
v=float(input())
p=m*v
print(f"{p:.2f}kgm/s")
```

Terminal Test cases

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1.1.2. Conditional Calculation Based on the Number of Digits

07:41

Write a Python program that accepts an integer n as input. Depending on the number of digits in n.

Constraints:
$1 \leq n \leq 999$

Input Format:
- The input consists of a single integer n.

Output Format:
- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases +

condition... Submit

```python
#write your code here...
n=int(input())
if 1<=n<10:
    print(n*n)
elif 10<=n<100:
    print(f"{n**(1/2):.2f}")
elif 100<=n<1000:
    print(f"{n**(1/3):.2f}")
else:
    print("Invalid")
```

Terminal Test cases

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1.1.3. Days between Two Dates

33:58

Write a Python program to calculate number of days between two dates.

Input Format:
- The first line contains the first date in the format $YYYY-MM-DD$.
- The second line contains the second date in the format $YYYY-MM-DD$.

Output Format:
- An integer representing the number of days between the given dates.

Note:
- The first date should always be considered earlier than the second date.

Sample Test Cases +

noOfDay... Submit

```python
#from datetime import date
from datetime import datetime

date1=input().strip()
date2=input().strip()

d1=datetime.strptime(date1,"%Y-%m-%d")
d2=datetime.strptime(date2,"%Y-%m-%d")

diff=d2-d1

print(diff.days)
```

Terminal Test cases

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1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

```python
#write your code here...
n=int(input())
s=str(n)
print(s[::-1])
```

Terminal Test cases

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1.1.5. Multiplication Table

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

```
<number> x <multiplier> = <result>
```

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

multiplica...

```python
#write your code here...
n=int(input())
for i in range(1,11):
    print(f"{n} x {i} = {n*i}")
```

Terminal Test cases

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1.2.1. Pass or Fail

Write a Python program that accepts the number of courses and the marks of a student in those courses.
The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n, the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses

passorFa...

```python

def cal(marks,courses):
    if any(mark < 40 for mark in marks):
        return "Fail"
    total_marks=sum(marks)
    score=(total_marks/(courses*100))*100

    if score>75:
        grade="Distinction"
    elif 60<=score<75:
        grade="First Division"
    elif 50<=score<60:
        grade="Second Division"
    elif 40<=score<50:
        grade="Third Division"
    return(score,grade)

courses=int(input())
marks=list(map(int,input().split()))
result= cal(marks,courses)

if result == "Fail":
    print("Fail")
else:
    score,grade=result
```

Terminal Test cases

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1.2.2. Fibonacci Series

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases +

fibonacci....

```python
#def fibonacci(n):
def fibonacci(n,a=0,b=1):
    if n==1:
        return a
    else:
        return fibonacci(n-1,b,a+b)


n = int(input())
for i in range(1, n + 1):
    print(fibonacci(i), end=" ")
```

Terminal Test cases

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1.2.3. Pattern - 1

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases +

rightangl...

```python
#Type Content here...
n=int(input())
for i in range(1,n+1):
    print('* '*i)
```

Terminal Test cases

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1.2.4. Pattern - 2

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Sample Test Cases +

numberP...

```python
rows=int(input())
for i in range(1,rows+1):
    for j in range(1,i+1):
        print(j,end=" ")
    print()
```

Terminal Test cases